Business Problem

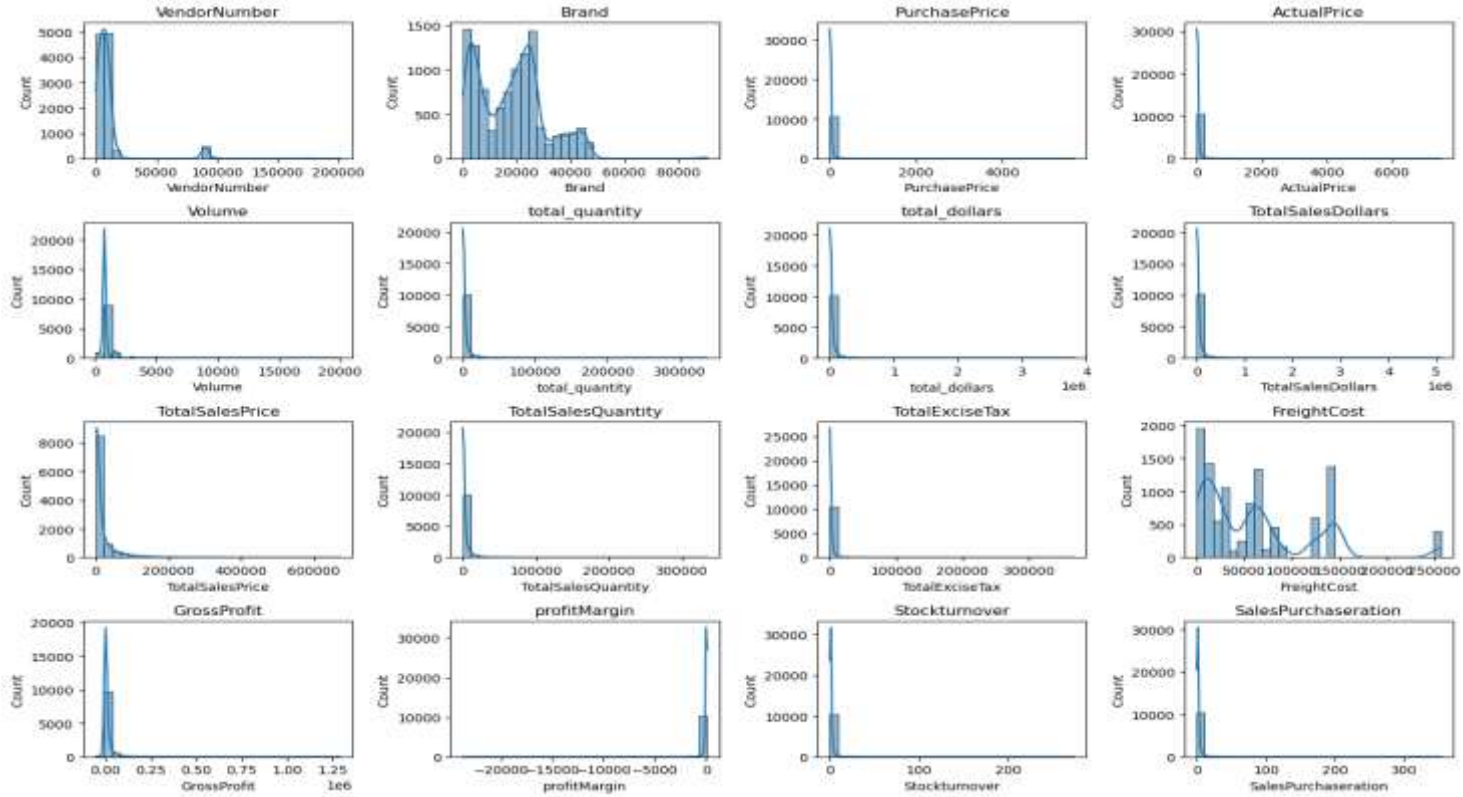
Effective inventory and sales management are critical for optimizing profitability in the retail and wholesale industry. Companies need to ensure that they are not incurring losses due to inefficient pricing, poor inventory turnover, or vendor dependency. The goal of this analysis is to:

- **Identify underperforming brands** that require promotional or pricing adjustments.
- **Determine top vendors** contributing to sales and gross profit.
- **Analyse the impact** of bulk purchasing on unit costs.
- **Assess inventory turnover** to reduce holding costs and improve efficiency.
- **Investigate the profitability variance** between high-performing and low-performing vendors.

Explanatory data analysis Insights

Summary statistics

	count	mean	std	min	25%	50%	75%	max
VendorNumber	10692.0	1.065065e+04	18753.519148	2.00	3951.000000	7153.000000	9552.000000	2.013590e+05
Brand	10692.0	1.803923e+04	12662.187074	58.00	5793.500000	18761.500000	25514.250000	9.063100e+04
PurchasePrice	10692.0	2.438530e+01	109.269375	0.36	6.840000	10.455000	19.482500	5.681810e+03
ActualPrice	10692.0	3.564367e+01	148.246016	0.49	10.990000	15.990000	28.990000	7.499990e+03
Volume	10692.0	8.473605e+02	664.309212	50.00	750.000000	750.000000	750.000000	2.000000e+04
total_quantity	10692.0	3.140887e+03	11095.086769	1.00	36.000000	262.000000	1975.750000	3.376600e+05
total_dollars	10692.0	3.010669e+04	123067.799627	0.71	453.457500	3655.465000	20738.245000	3.811252e+06
TotalSalesDollars	10692.0	4.223907e+04	167655.265984	0.00	729.220000	5298.045000	28396.915000	5.101920e+06
TotalSalesPrice	10692.0	1.879378e+04	44952.773386	0.00	289.710000	2857.800000	16059.562500	6.728193e+05
TotalSalesQuantity	10692.0	3.077482e+03	10952.851391	0.00	33.000000	261.000000	1929.250000	3.349390e+05
TotalExciseTax	10692.0	1.774226e+03	10975.582240	0.00	4.800000	46.570000	418.650000	3.682428e+05
FreightCost	10692.0	6.143376e+04	60938.458032	0.09	14069.870000	50293.620000	79528.990000	2.570321e+05
GrossProfit	10692.0	1.213238e+04	46224.337964	-52002.78	52.920000	1399.640000	8660.200000	1.290668e+06
profitMargin	10692.0	-inf	NaN	-inf	13.324515	30.405457	39.956135	9.971666e+01
Stockturnover	10692.0	1.706793e+00	6.020460	0.00	0.807229	0.981529	1.039342	2.745000e+02
SalesPurchaseration	10692.0	2.504390e+00	8.459067	0.00	1.153729	1.436894	1.665449	3.529286e+02


VendorNumber
Count
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ActualPrice
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ActualPrice
Volume
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Volume
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total_quantity
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TotalSalesDollars
Count
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TotalSalesPrice
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TotalSalesPrice
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TotalSalesQuantity
TotalExciseTax
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FreightCost
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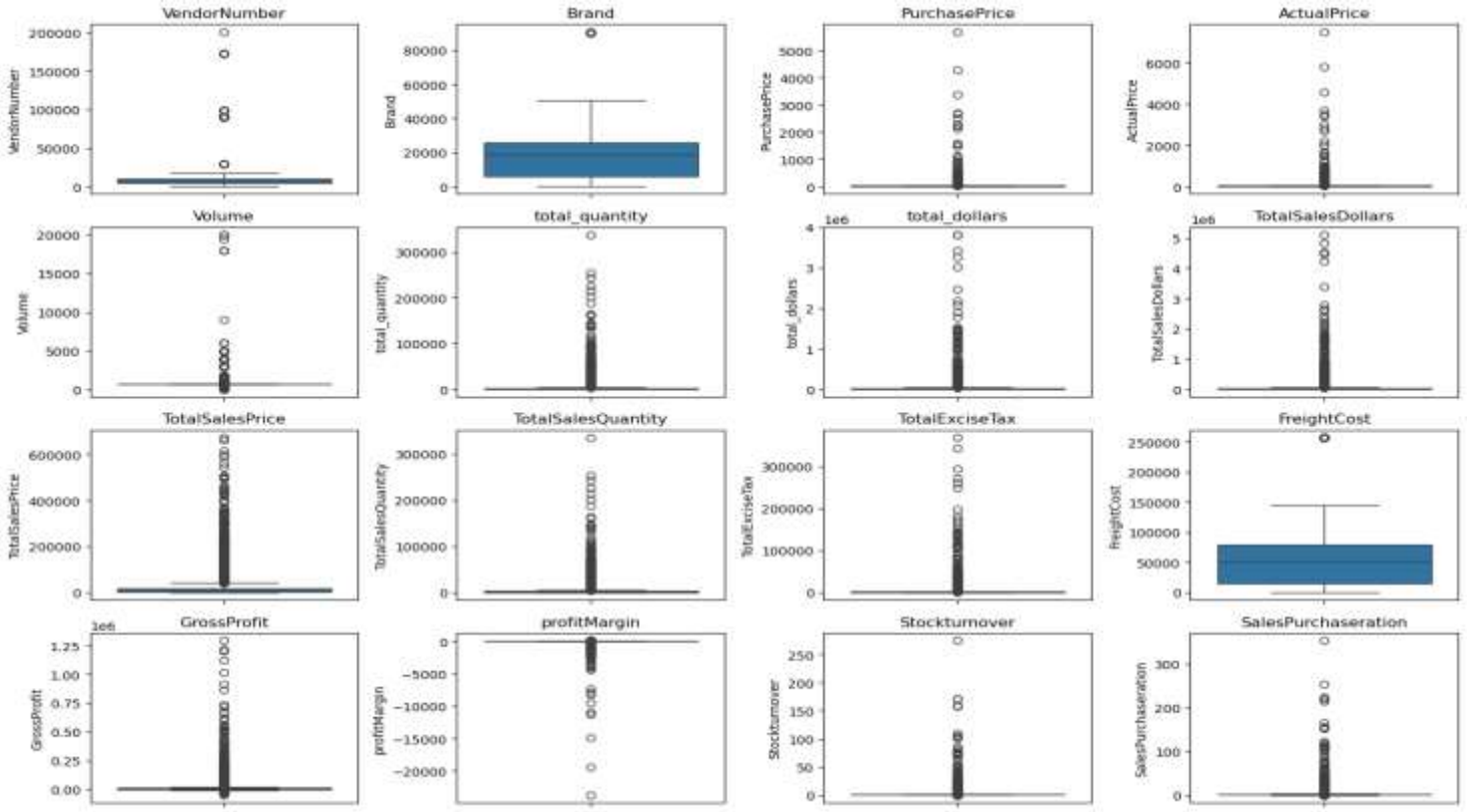
VendorNumber
VendorNumber
Brand
Brand
PurchasePrice
PurchasePrice
ActualPrice
ActualPrice
Volume
Volume
total_quantity
total_quantity
total_dollars
1e6
total_dollars
TotalSalesDollars
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TotalSalesDollars
TotalSalesPrice
TotalSalesPrice
TotalSalesQuantity
TotalSalesQuantity
TotalExciseTax
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FreightCost
FreightCost
GrossProfit
1e6
GrossProfit
profitMargin
profitMargin
Stockturnover
Stockturnover
SalesPurchaseration
SalesPurchaseration

Negatives and zeros:

- **Gross Profit:** Minimum of **(-52,0022.78)**, Indicating potential losses due to high costs or heavy discounts. This could be due to selling products at lower prices than their purchase costs.
- **Profit Margin :** Has a minimum of **(-inf)**, Which suggests an instances Where revenue is zero or even lower than the total cost Leading to extreme negative profit margins.
- **Total sales quantity and sales dollars:** Some products **show zero sales** indicating that they were never purchased But never sold. These may be slow moving or obstacles obsolete stock, Leading to inventory Inefficiencies.

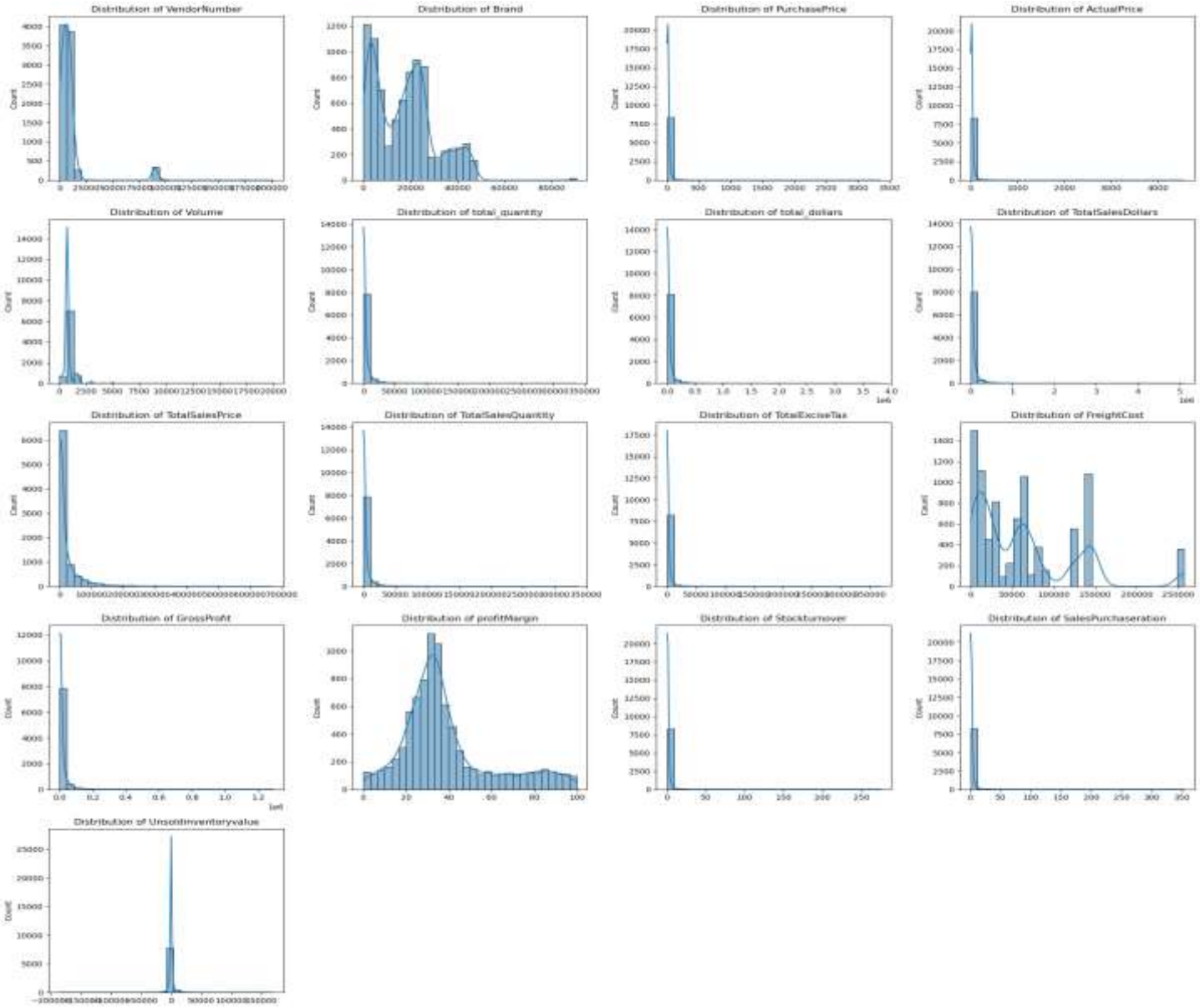
Outliers detected by high standard deviations:

- Purchase price and actual selling price have very high maximums **(around 5,681.81 and 7,499.99)** relative to their means, indicating a small set of premium or niche products priced far above the core range.
- Freight cost ranges from **0.09 to 257,032.07**, revealing extreme variation in logistics costs that may stem from bulk shipments, long-distance suppliers, or inefficient freight allocation.
- Stock turnover ranges from **0 to 274.5**, meaning some products barely move while others sell extremely fast; values above 1 imply sales are drawing on older inventory or prior purchase cycles rather than only current-period purchases.

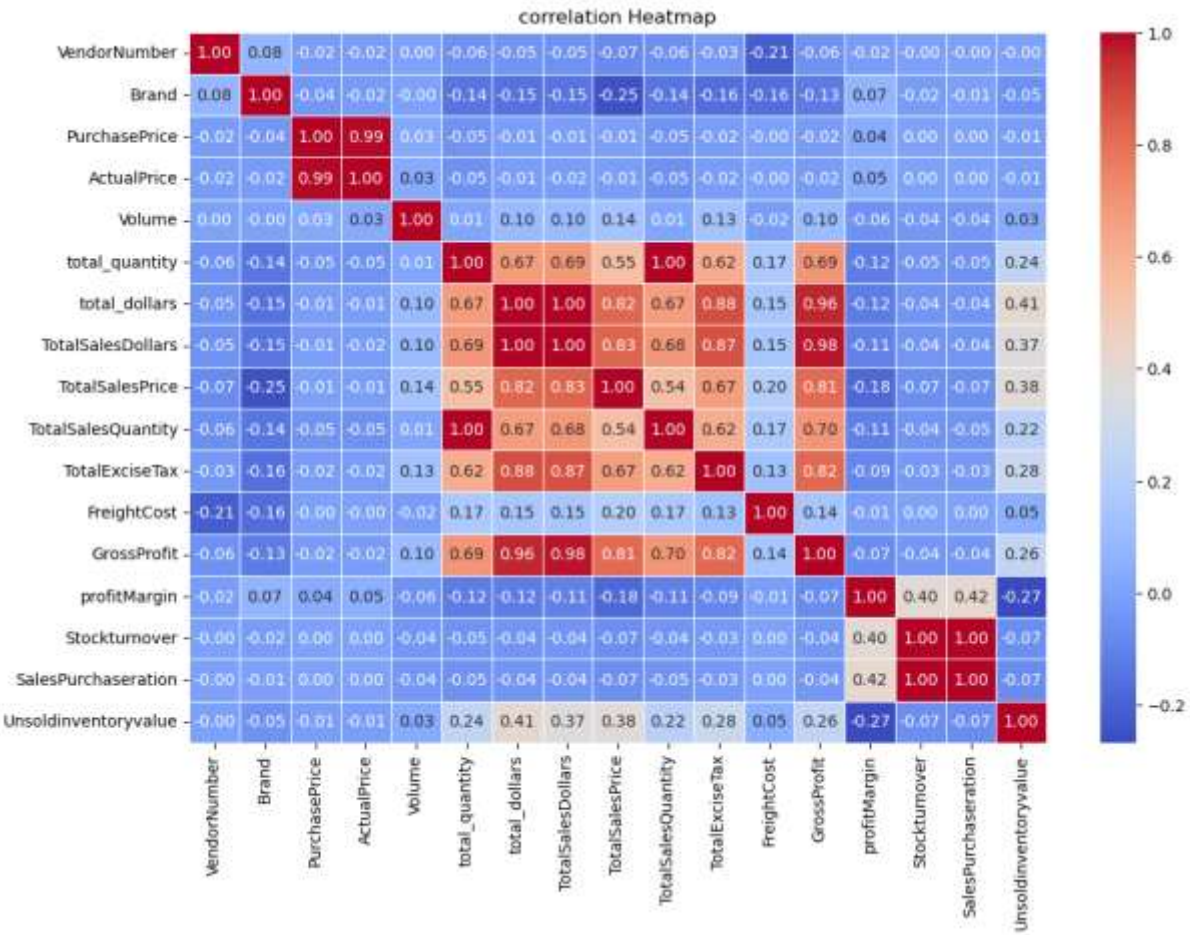
Data Filtering

To enhance the readability of the insights we remove the inconsistency data

- **Gross profit <= 0 (To exclude the transactions leading to losses).**
- **Profit margin <= 0 (To ensure analysis focuses on profitability transactions).**
- **Total Sales <= 0 (To eliminate inventory that was never sold).**


Distribution of VendorNumber
Distribution of Brand
Distribution of PurchasePrice
Distribution of ActualPrice
Distribution of Volume
Distribution of total_quantity
Distribution of total_dollars
Distribution of TotalSalesDollars
Distribution of TotalSalesPrice
Distribution of TotalSalesQuantity
Distribution of TotalExciseTax
Distribution of FreightCost
Distribution of GrossProfit
Distribution of profitMargin
Distribution of Stockturnover
Distribution of SalesPurchaseratio
Distribution of Unsoldinventoryvalue
Count

Correlation Insights



- Purchase price has a weak correlation with the total sales dollars And gross profit Suggesting that the price variation do not significantly impact sales revenue or profit
- Strong correlation between total purchase quantity and total sales quantity confirming efficient inventory turnover.
- Negative correlation between profit Margin And total sales price Suggest that that as sales price increases margin decreases possibly due to competitive price impression
- Stock turnover has a weak negative correlation with both gross profit and profit margin Indicating that faster than our does not necessarily result in higher profitability

Research questions and key findings

1. Brands for promotional or pricing adjustments

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                                TotalSalesDollars  profitMargin
Description
Santa Rita Organic Svgn Bl                   9.99     66.466466
Debauchery Pnt Nr                           11.58     65.975820
Concannon Glen Ellen Wh Zin                 15.95     83.448276
Crown Royal Apple                           27.86     89.806174
Sauza Sprklg Wild Berry Marg                27.96     82.153076
...                                           ...           ...
Sbragia Home Ranch Merlot                  549.75     66.444748
Smirnoff Fluffed Marshmallow               554.63     61.392280
Vino.e Falanghina Beneventno               558.57     63.352847
Goulee Cos d'Estournel 10                  558.87     69.434752
Atlas Peak Cab Svgn                        559.84     60.803087

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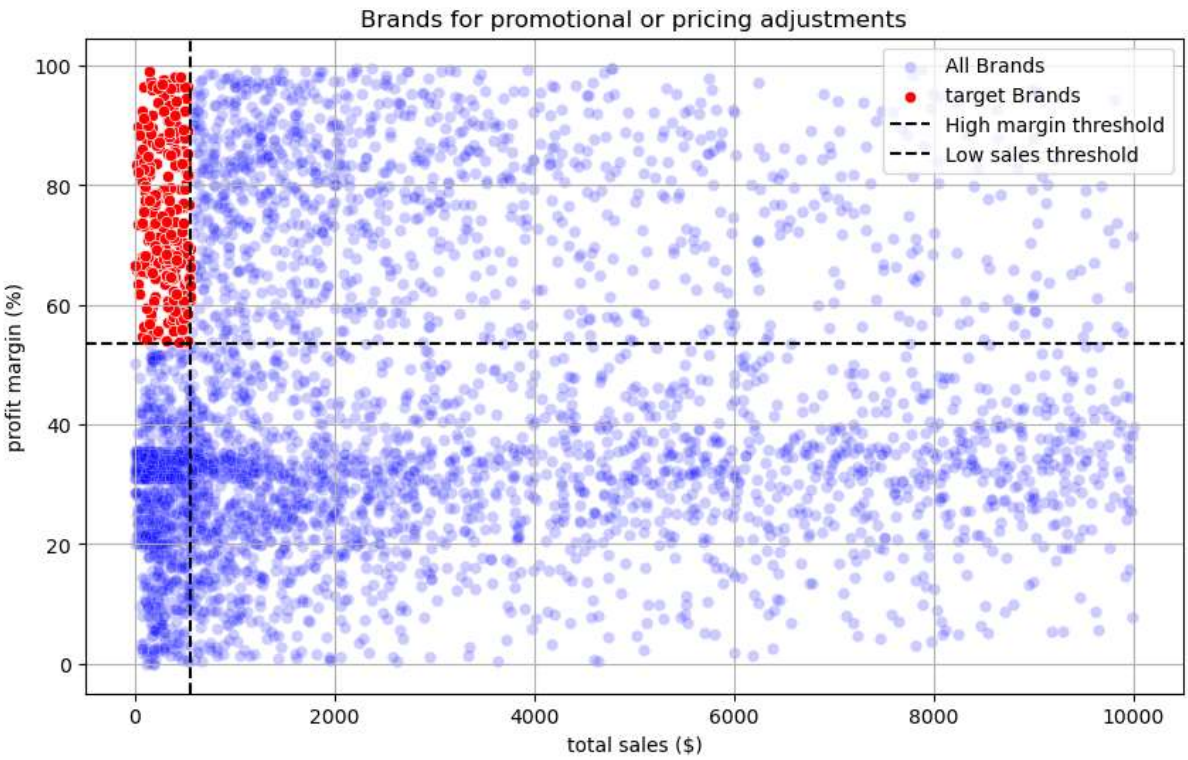
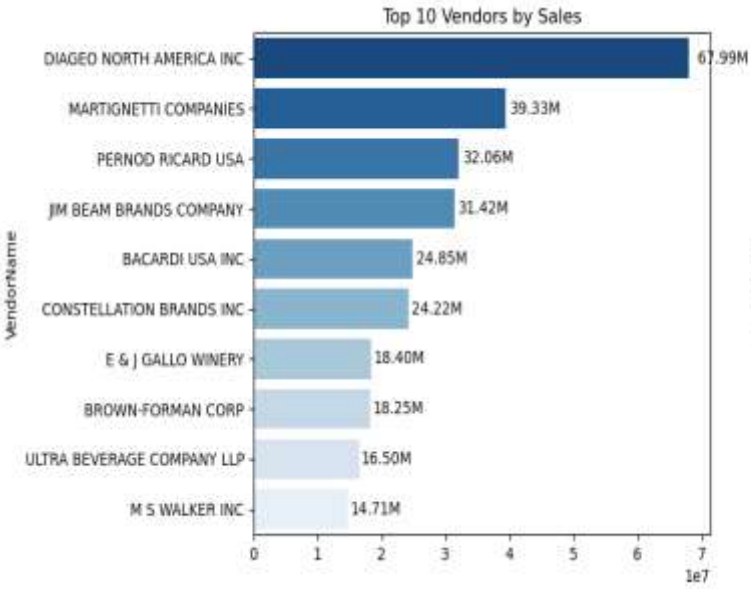
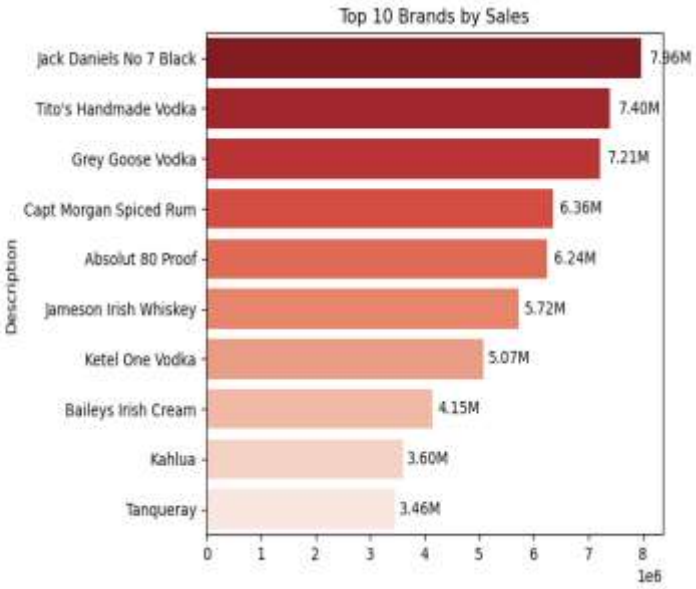
Total 216 brands exhibits lower sales but higher profit margins which could benefit from targeted marketing's promotions or price optimizations to increase volume without compromising profitability

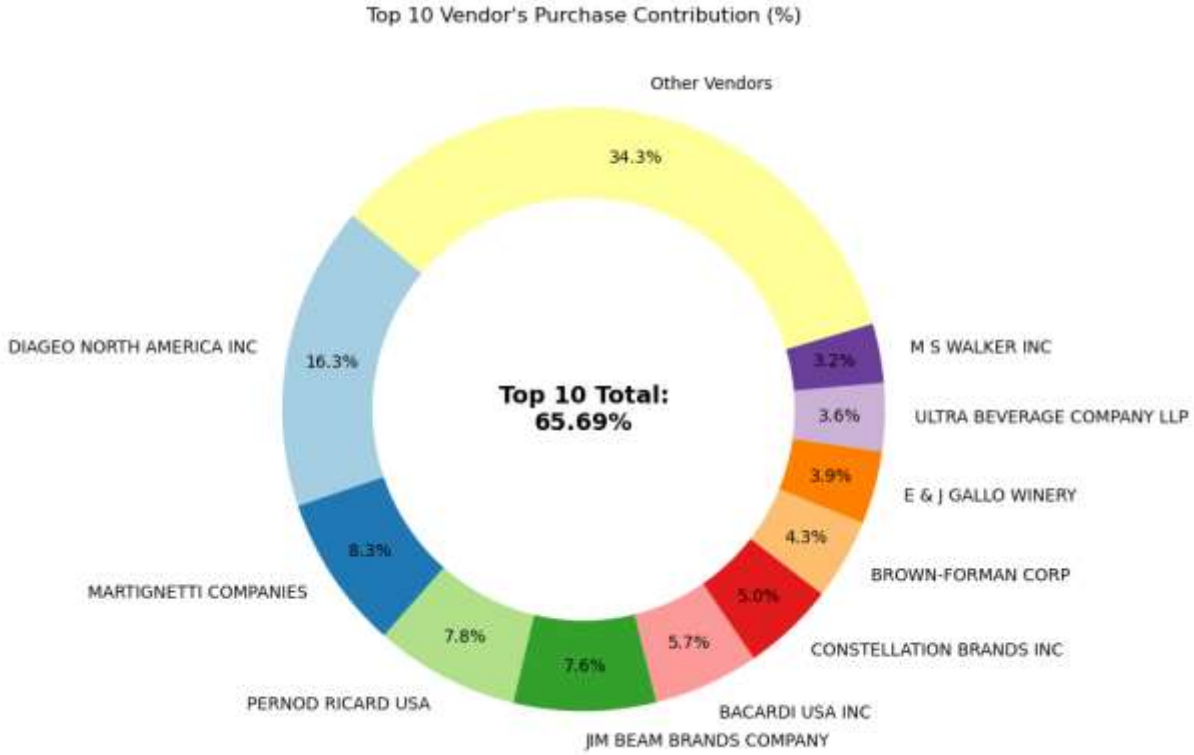


2. Which brands and vendors demonstrate the highest sales performance





3. Top vendors buy sales and purchase contribution
   The most top 10 vendors contribution is 65.69 % of total purchases, while the Remaining vendors contribute only 34.31 % that over reliance on a few vendors may introduce risks such as supply chain disruptions indicating a need for diversification.



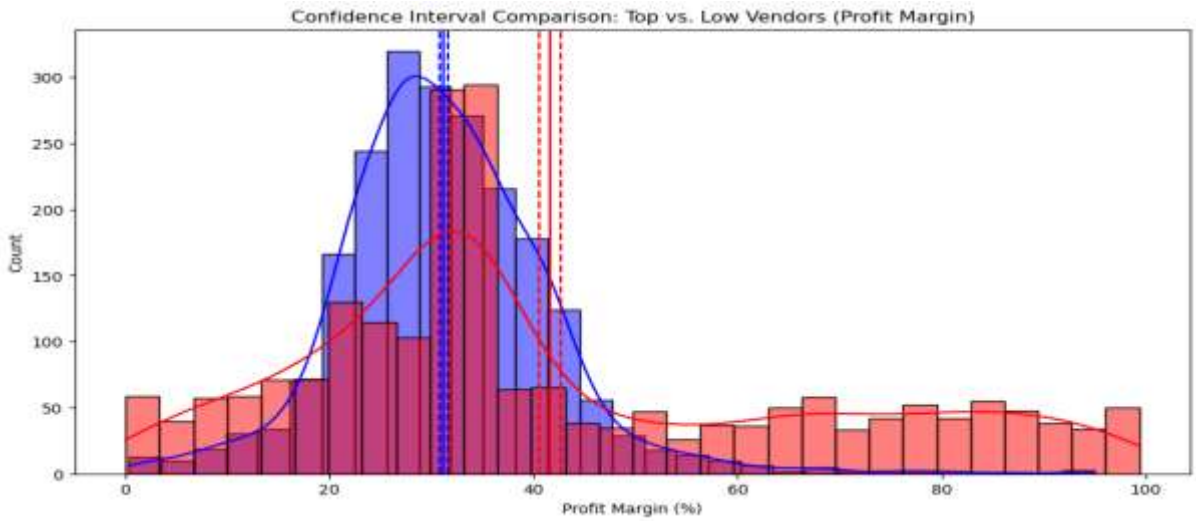
4. Which vendors have low inventory turnover indicating excess stock and slow moving products? how much capital is locked in unsold inventory per vendor and which vendors contribute the most to it?

The total unsold inventory capital is $ 2.71M
Slow Moving inventory increases storage costs then reduces the cash flow and affects the overall profitability
Identifying the vendors with low inventory turnover enables better stock management and minimising the financial strain

VendorName	Stockturnover
ALISA CARR BEVERAGES	0.615385
HIGHLAND WINE MERCHANTS LLC	0.708333
PARK STREET IMPORTS LLC	0.751306
Circa Wines	0.755676
Dunn Wine Brokers	0.766022
CENTEUR IMPORTS LLC	0.773953
SMOKY QUARTZ DISTILLERY LLC	0.783835
TAMWORTH DISTILLING	0.797078
THE IMPORTED GRAPE LLC	0.807569
WALPOLE MTN VIEW WINERY	0.820548

	VendorName	Unsoldinventoryvalue
25	DIAGEO NORTH AMERICA INC	722.21K
46	JIM BEAM BRANDS COMPANY	554.67K
68	PERNOD RICARD USA	470.63K
116	WILLIAM GRANT & SONS INC	401.96K
30	E & J GALLO WINERY	228.28K
79	SAZERAC CO INC	198.44K
11	BROWN-FORMAN CORP	177.73K
20	CONSTELLATION BRANDS INC	133.62K
61	MOET HENNESSY USA INC	126.48K
77	REMY COINTREAU USA INC	118.60K

5. What is 95% confidence interval for profit margins of top performings and low performing vendors



Top Vendors 95% CI: (30.74, 31.61), Mean: 31.18

Low Vendors 95% CI: (40.50, 42.64), Mean: 41.57

- This suggests that the vendors with lower sales trend Maintain higher profit margin potentially due to the premium pricing or low operational costs
- For high performance vendor if they aim to improve profitability they could explore more selective price adjustments cost optimization or bonding strategies
- For the low performance vendor Despite higher margins Their low sales volume might indicate A need for Better marketing or Pricing adjustments are Improved distributions Or promoting their products In a better way

Is there a significant difference in profit margins between top performing and low performing vendors

Hypothesis :

H0 (Null hypothesis) : There is no significant difference in the mean profit margins of top performing and do performing vendors

H1 (Alternative hypothesis) : The main profit margins of top performing and low performing vendors are significantly different

t-Statistic: -17.6695

p-value: 2.2969e-66

Conclusion: There is a SIGNIFICANT difference in profit margins between top and low performing vendors.

Decision: Rejecting H0 (Null Hypothesis)